

Chapter- 4

4. Network theories and Equivalent circuits

4. Network theories and Equivalent circuits

- Kirchhoff's Rules
- Mesh analysis
- Thevenin's Theorem
- Norton's Theorem
- Δ -Y and Y- Δ Conversions

Network theories (theorems)

- **Network theorems** are rules that help us analyze complex electrical circuits by reducing them into simpler forms without changing their electrical behavior.

Kirchhoff's Rules

- Combinations of resistors can be simplified and analyzed using the expression $\Delta V = IR$ and the rules for series and parallel combinations of resistors. Very often, however, it is not possible to reduce a circuit to a single loop.
- The procedure for analyzing more complex circuits is made possible by using the **two** following principles, called **Kirchhoff's rules**.

1. **Junction rule.** At any junction, the sum of the currents must equal zero:

$$\sum_{\text{junction}} I = 0$$



2. **Loop rule.** The sum of the potential differences across all elements around any closed circuit loop must be zero:

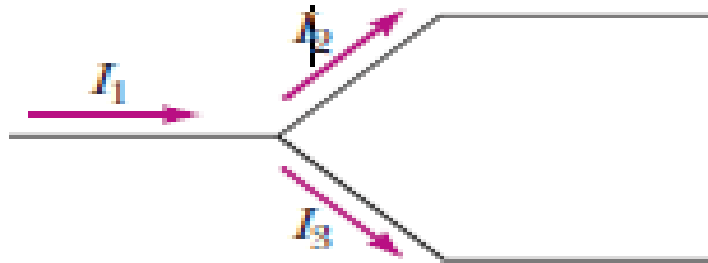
$$\sum_{\text{closed loop}} \Delta V = 0$$

Network theories and Equivalent circuits,
Wubishet G, UU, 2026



Cont.

- **Kirchhoff's first rule** is a statement of **conservation of electric charge**. All charges that enter a given point in a circuit must leave that point because charge cannot build up at a point.



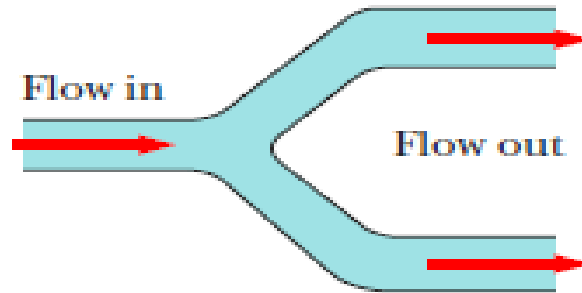
- Currents directed into the junction are entered into the junction rule as **+/**, whereas currents directed out of a junction are entered as **-/**.

Cont.

- Applying this rule to the junction:-

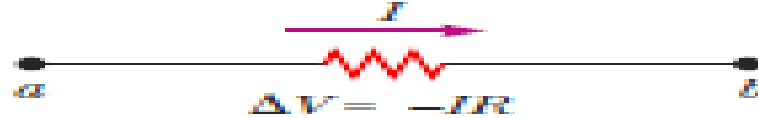
$$I_1 - I_2 - I_3 = 0$$

- Kirchhoff's second** rule follows from the law of **conservation of energy**.

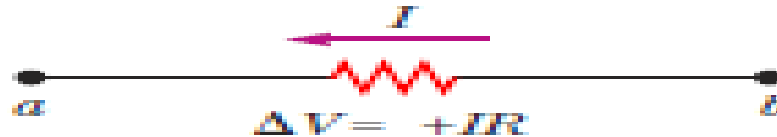


- The following sign conventions apply when using the second rule. Rules for determining the potential differences across a resistor and a battery:-

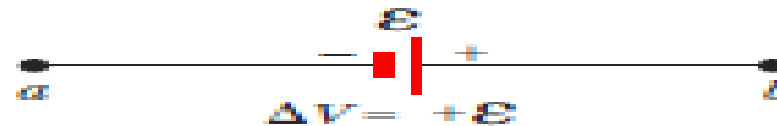
- ❖ Charges move from the high-potential end of a resistor toward the low potential end, so if a resistor is traversed in the direction of the current, the potential difference ΔV across the resistor is $-IR$.



- ❖ If a resistor is traversed in the direction *opposite* the current, the potential difference ΔV across the resistor is $+IR$.



- ❖ If a source of emf is traversed in the direction of the emf (from negative to positive), the potential difference ΔV is $+\mathcal{E}$.



- ❖ If a source of emf is traversed in the direction opposite the emf (from positive to negative), the potential difference ΔV is $-\mathcal{E}$.

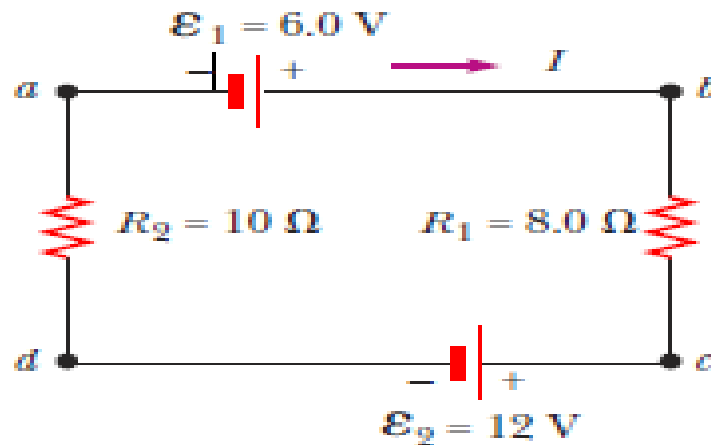


Cont.

- In general, to solve a particular circuit problem, the **number of independent equations** you need to obtain from the **two rules** equals **the number of unknown currents**.

Cont.

- **Example 1**:- A single-loop circuit contains two resistors and two batteries as shown in figure. Find the current in the circuit.



Apply Kirchhoff's loop rule to the single loop in the circuit:

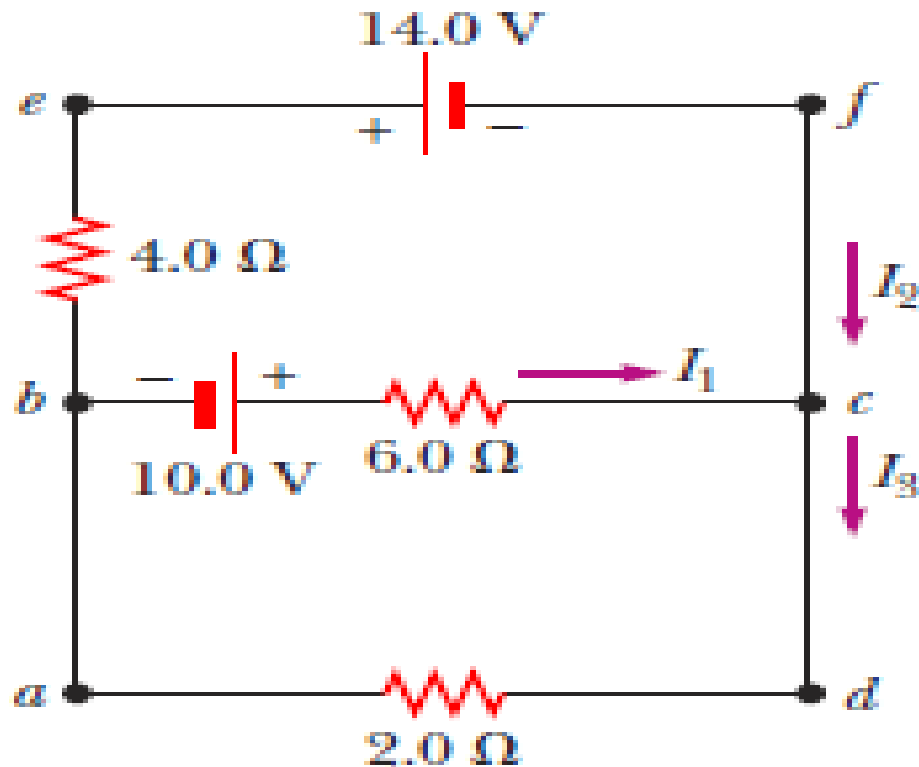
$$\sum \Delta V = 0 \rightarrow \mathcal{E}_1 - IR_1 - \mathcal{E}_2 - IR_2 = 0$$

Solve for I and use the values given in Figure 28.14:

$$(1) \quad I = \frac{\mathcal{E}_1 - \mathcal{E}_2}{R_1 + R_2} = \frac{6.0 \text{ V} - 12 \text{ V}}{8.0 \Omega + 10 \Omega} = -0.33 \text{ A}$$

Cont.

- **Example 2**:-Find the currents I_1 , I_2 and I_3 in the circuit shown in figure below.



Apply Kirchhoff's junction rule to junction c :

$$(1) \quad I_1 + I_2 - I_3 = 0$$

We now have one equation with three unknowns: I_1 , I_2 , and I_3 . There are three loops in the circuit: $abcda$, $befcb$, and $aefda$. We need only two loop equations to determine the unknown currents. (The third loop equation would give no new information.) Let's choose to traverse these loops in the clockwise direction. Apply Kirchhoff's loop rule to loops $abcda$ and $befcb$:

$$abcda: (2) \quad 10.0 \text{ V} - (6.0 \, \Omega)I_1 - (2.0 \, \Omega)I_3 = 0$$

$$befcb: -(4.0 \, \Omega)I_2 - 14.0 \text{ V} + (6.0 \, \Omega)I_1 - 10.0 \text{ V} = 0$$

$$(3) \quad -24.0 \text{ V} + (6.0 \, \Omega)I_1 - (4.0 \, \Omega)I_2 = 0$$

Solve Equation (1) for I_3 and substitute into Equation (2):

$$10.0 \text{ V} - (6.0 \, \Omega)I_1 - (2.0 \, \Omega)(I_1 + I_2) = 0$$

$$(4) \quad 10.0 \text{ V} - (8.0 \, \Omega)I_1 - (2.0 \, \Omega)I_2 = 0$$

Multiply each term in Equation (3) by 4 and each term in Equation (4) by 3:

$$(5) \quad -96.0 \text{ V} + (24.0 \, \Omega)I_1 - (16.0 \, \Omega)I_2 = 0$$

$$(6) \quad 30.0 \text{ V} - (24.0 \, \Omega)I_1 - (6.0 \, \Omega)I_2 = 0$$

Add Equation (6) to Equation (5) to eliminate I_1 and find I_2 :

$$-66.0 \text{ V} - (22.0 \, \Omega)I_2 = 0$$

$$I_2 = -3.0 \text{ A}$$

Use this value of I_2 in Equation (3) to find I_1 :

$$-24.0 \text{ V} + (6.0 \, \Omega)I_1 - (4.0 \, \Omega)(-3.0 \text{ A}) = 0$$

$$-24.0 \text{ V} + (6.0 \, \Omega)I_1 + 12.0 \text{ V} = 0$$

$$I_1 = 2.0 \text{ A}$$

Use Equation (1) to find I_3 :

Network theories and Equivalent circuits,
Wubishet G, UU, 2026

$$I_3 = I_1 + I_2 = 2.0 \text{ A} - 3.0 \text{ A} = -1.0 \text{ A}$$

Mesh Analysis

- The method that is used to determine the current flowing around a mesh or loop in the closed complex circuits is known as **mesh analysis**.
- A loop is a closed path with no node passed more than once.
- **A mesh** is a loop that does not contain any other loop within it.
- A **branch** is a path that joins two nodes and it contains a circuit element.
- If a branch belongs to **only one mesh**, then the **branch current will be equal to mesh current**.
- If a branch is common to two meshes, then the **branch current will be equal to the sum (or difference) of two mesh currents**, when they are in same (or opposite) direction.

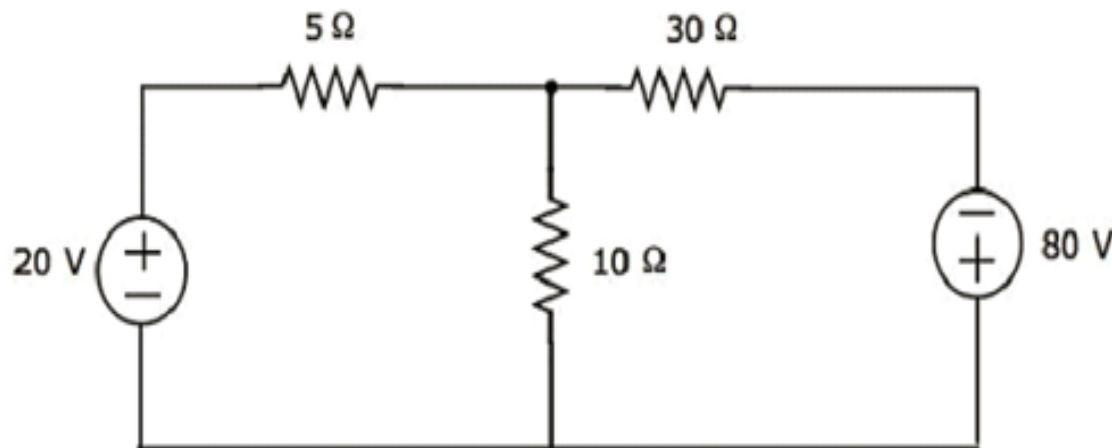
Procedure of Mesh Analysis

Follow these steps while solving any electrical network or circuit using Mesh analysis.

1. Identify the **meshes** and **label the mesh currents** in either clockwise or anti-clockwise direction.
2. Observe the **amount of current** that flows through each element in terms of **mesh currents**.
3. Write **mesh equations** to all meshes. Mesh equation is obtained by applying **KVL** first and then **Ohm's law**.
4. Solve the mesh equations obtained in Step 3 in order to get the **mesh currents**.

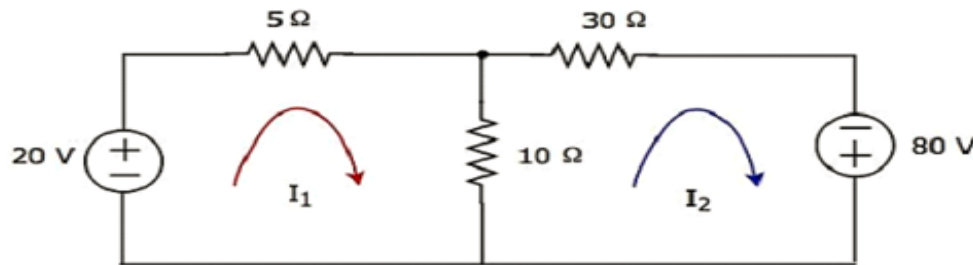
Cont.

- **Example**:- Using mesh analysis, determine the current across each resistor.



Cont.

- **Step 1** – There are two meshes in the above circuit. The **mesh currents** I_1 and I_2 are considered in clockwise direction.



- **Step 2** – The mesh current I_1 flows through 20 V voltage source and 5 Ω resistor. Similarly, the mesh current I_2 flows through 30 Ω resistor and -80 V voltage source. But, the difference of two mesh currents, I_1 and I_2 , flows through 10 Ω resistor, since it is the common branch of two meshes.
- **Step 3** – In this case, we will get **two mesh equations** since there are two meshes in the given circuit. Write **mesh equations** to all meshes. Mesh equation is obtained by applying KVL first and then Ohm's law.

Cont.

The **mesh equation** of first mesh is The **mesh equation** of second mesh is

$$20 - 5I_1 - 10(I_1 - I_2) = 0$$

$$\Rightarrow 20 - 15I_1 + 10I_2 = 0$$

$$\Rightarrow 10I_2 = 15I_1 - 20$$

Divide the above equation with 5.

$$2I_2 = 3I_1 - 4$$

Multiply the above equation with 2.

$$4I_2 = 6I_1 - 8$$

$$-10(I_2 - I_1) - 30I_2 + 80 = 0$$

Divide the above equation with 10.

$$-(I_2 - I_1) - 3I_2 + 8 = 0$$

$$\Rightarrow -4I_2 + I_1 + 8 = 0$$

$$4I_2 = I_1 + 8$$

Cont.

- **Step 4** – Finding mesh currents I_1 and I_2 by solving Equation 1 and Equation 2.

$$6I_1 - 8 = I_1 + 8$$

$$\Rightarrow 5I_1 = 16$$

$$\Rightarrow I_1 = \frac{16}{5} A$$

Substitute I_1 value in Equation 2.

$$4I_2 = \frac{16}{5} + 8$$

$$\Rightarrow 4I_2 = \frac{56}{5}$$

$$\Rightarrow I_2 = \frac{14}{5} A$$

Thevenin's Theorem

Thevenin's Theorem state that:

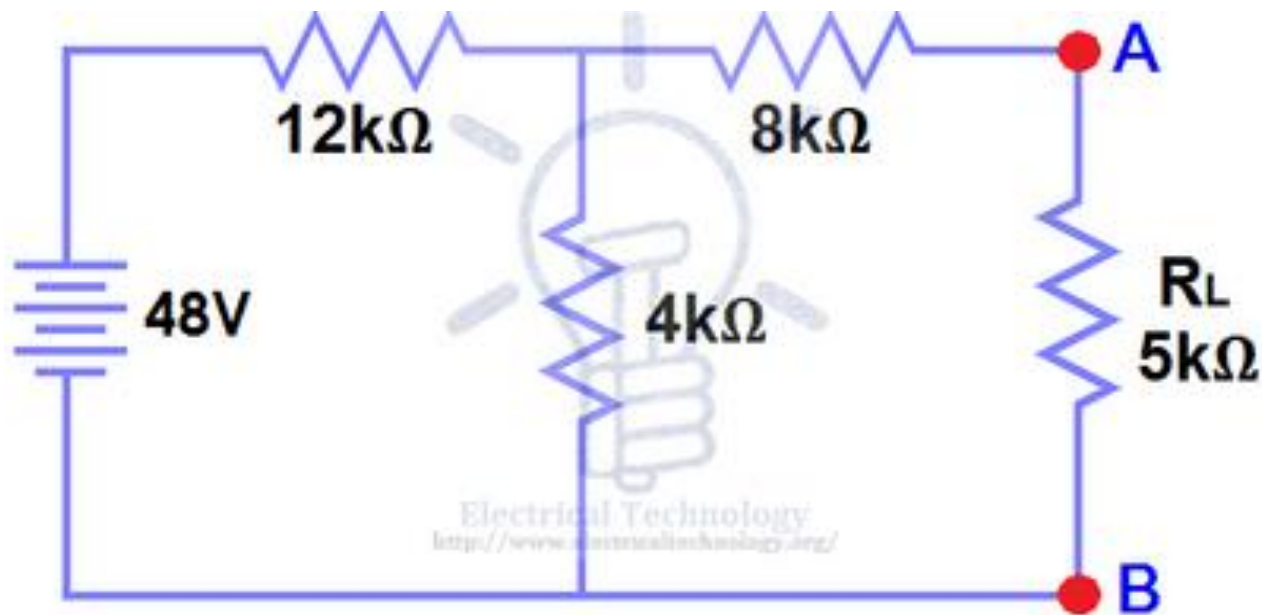
- Any complex circuit with current and voltage sources can be replaced by an equivalent circuit containing a single independent voltage source V_{TH} and a Series Resistance R_{TH} .
- V_{TH} = Thevenin's Voltage
- R_{TH} = Thevenin's Resistance

Steps to Analyze an Electric Circuit using Thevenin's Theorem

1. Open the load resistor. Calculate / measure the open circuit voltage. This is the **Thevenin Voltage (V_{TH})**.
2. Open current sources and short voltage sources. Calculate /measure the Open Circuit Resistance. This is the **Thevenin Resistance (R_{TH})**.
3. Draw the circuit with measured V_{TH} and measured R_{TH} as a **series resistance** and connect the load resistor which we had removed in Step (1). This is the **equivalent Thevenin circuit**.
4. Now find the **Total current** flowing through the load resistor by using the **Ohm's Law** : $I_T = V_{TH} / (R_{TH} + R_L)$.

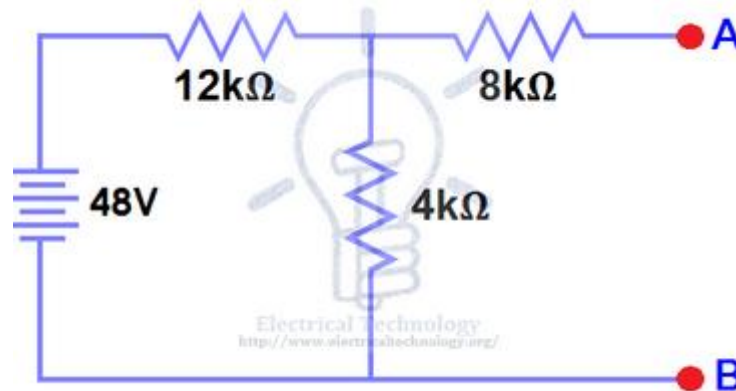
Solved Example by Thevenin's Theorem:

1. Find V_{TH} , R_{TH} and the load current I_L flowing through and load voltage across the load resistor in fig below *by using Thevenin's Theorem*.



Cont.

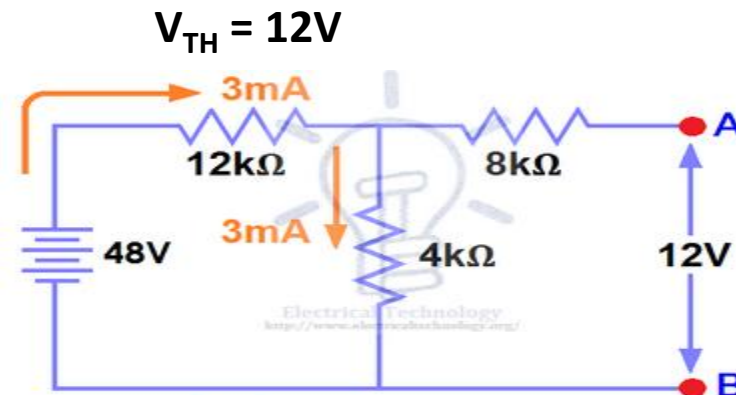
- **STEP 1.** Open the $5\text{k}\Omega$ load resistor and Calculate / measure the open circuit voltage. This is the **Thevenin Voltage (V_{TH})**.



- Since **3mA** current flows in both **$12\text{k}\Omega$** and **$4\text{k}\Omega$** resistors as this is a series circuit and current will not flow in the **$8\text{k}\Omega$** resistor as it is open.

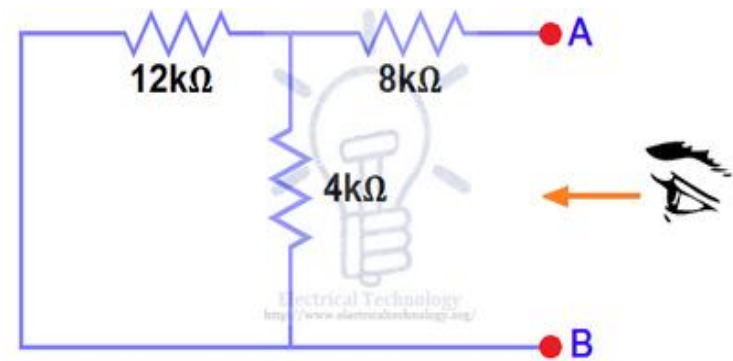
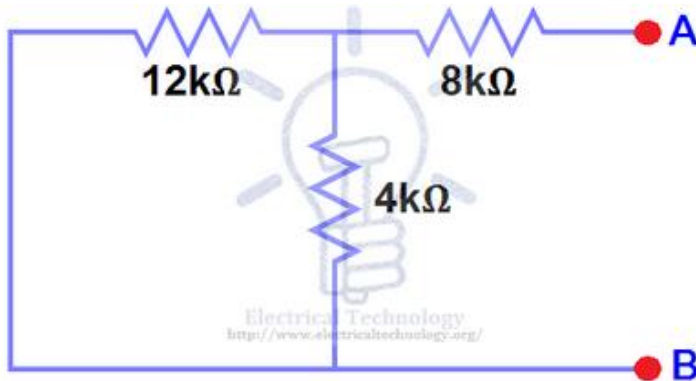
Cont.

- This way, **12V** (**3mA** x **4kΩ**) will appear across the **4kΩ resistor**. We also know that current is not flowing through **the 8kΩ resistor** as it is an open circuit, but the **8kΩ** resistor is in parallel with **4kΩ resistor**. So the same voltage i.e. **12V** will appear across the **8kΩ resistor** as well as **4kΩ resistor**. Therefore **12V will appear across the AB terminals**.



Cont.

- **STEP 2. Open current sources and short voltage sources.** Calculate / measure the open circuit resistance. This is the **Thevenin Resistance (R_{TH})**



- We can see that **8kΩ resistor** is in series with a parallel connection of **4kΩ resistor** and **12kΩ resistor**.

$8k\Omega + (4k\Omega \parallel 12k\Omega) \dots (\parallel = \text{in parallel with})$

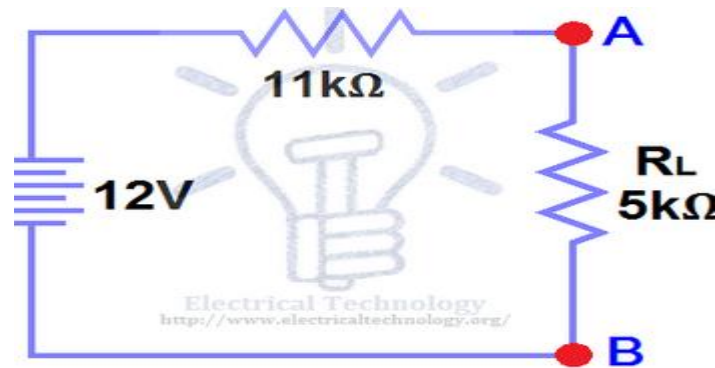
$$R_{TH} = 8k\Omega + [(4k\Omega \times 12k\Omega) / (4k\Omega + 12k\Omega)]$$

$$R_{TH} = 8k\Omega + 3k\Omega$$

$$R_{TH} = 11k\Omega$$

Cont.

- **STEP 3.** Connect the R_{TH} in series with Voltage Source V_{TH} and re-connect the load resistor. This is the **Thevenin's equivalent circuit**.



- **STEP 4.**
- Now apply **Ohm's law** . Calculate the total load current

$$I_L = V_{TH} / (R_{TH} + R_L)$$

$$I_L = 12V / (11k\Omega + 5k\Omega) \rightarrow = 12/16k\Omega$$

$$I_L = 0.75mA$$

Norton's Theorem

Norton's Theorem states that:

- Any complex circuit with current and voltage sources can be replaced by an equivalent circuit containing a **single independent current source I_N** and a **parallel resistance R_N** .

Steps to Analyze an Electric Circuit using Norton's Theorem

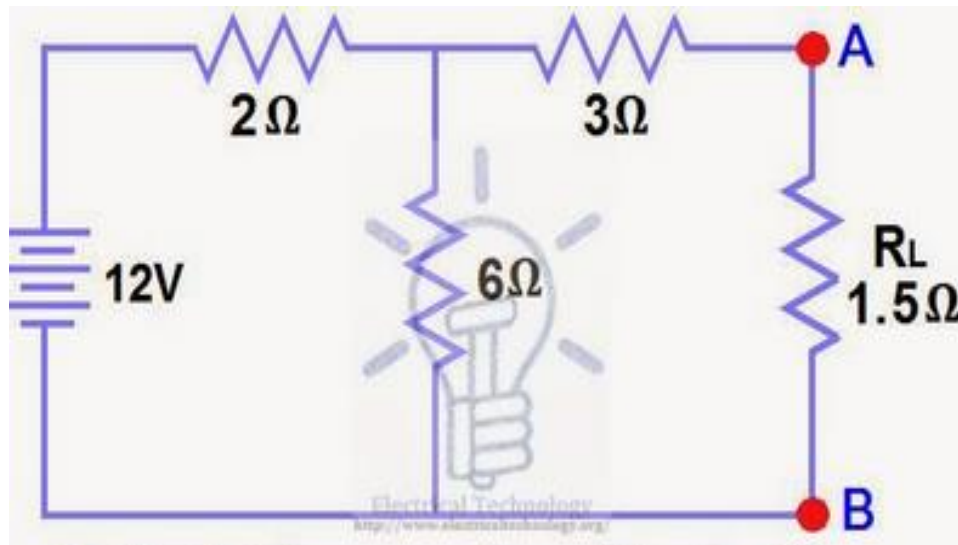
1. Short the load resistor. Calculate / measure the Short Circuit Current. This is the **Norton Current (I_N)**.
2. Open Current Sources, Short Voltage Sources and Open Load Resistor. Calculate /measure the Open Circuit Resistance. This is the **Norton Resistance (R_N)**.
3. Draw the circuit with measured **I_N and R_N** as a parallel resistance and connect the load resistor. This is the Equivalent Norton Circuit of Complex circuit.
4. Now find the Load current flowing through and Load Voltage across Load Resistor by using the **Current divider rule**.

$$I_L = I_N / (R_N / (R_N + R_L))$$

Network theories and Equivalent circuits,
Wubishet G, UU, 2026

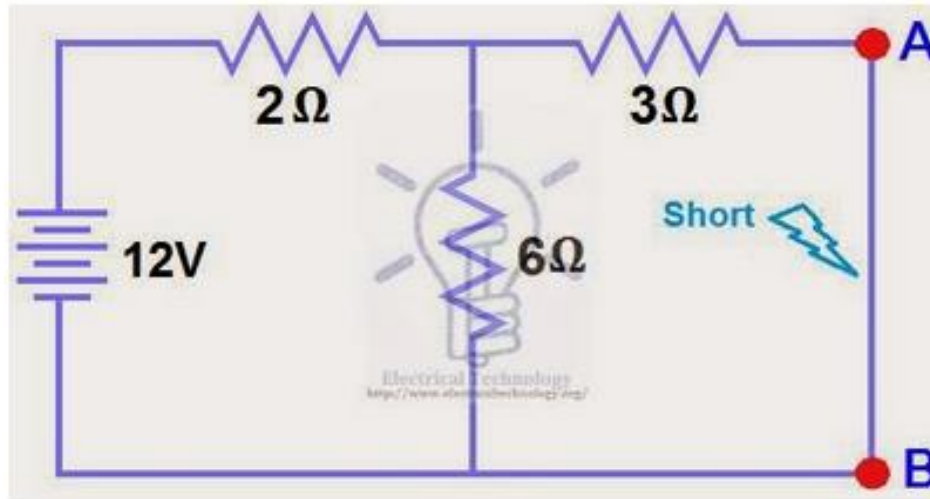
Solved Example by Norton's Theorem:

1. Find R_N , I_N , the current flowing through and Load Voltage across the load resistor in fig below by using Norton's Theorem.



Cont.

- **STEP 1. Short the 1.5Ω load resistor and Calculate / measure the Short Circuit Current.** This is the **Norton Current (I_N)**.



- The 6Ω and 3Ω are then in parallel and this parallel combination of 6Ω and 3Ω are then in **series** with 2Ω .

Cont.

- So the **total resistance** of the circuit to the Source is:-
 $2\Omega + (6\Omega \parallel 3\Omega) \dots (\parallel = \text{in parallel with})$.

$$R_T = 2\Omega + [(3\Omega \times 6\Omega) / (3\Omega + 6\Omega)]$$

$$R_T = 2\Omega + 2\Omega = 4\Omega.$$

$$R_T = 4\Omega$$

$$I_T = V \div R_T$$

$$I_T = 12V \div 4\Omega$$

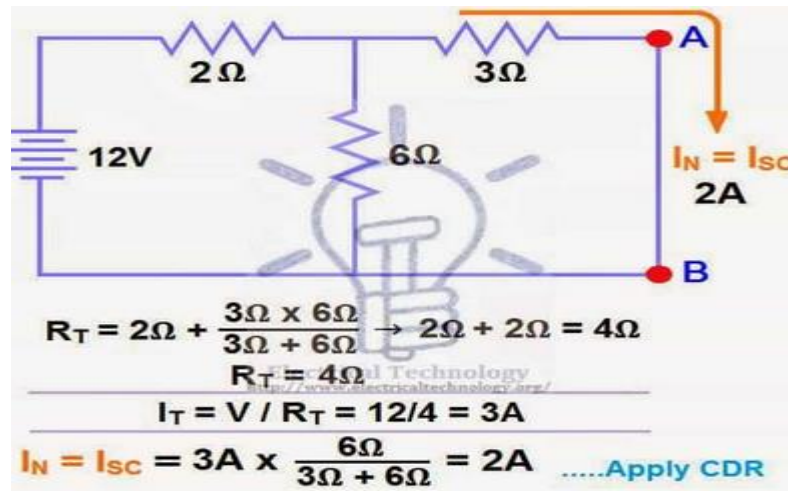
$$I_T = 3A.$$

- Now we have to find $I_{SC} = I_N$... Apply CDR...
(Current Divider Rule)...

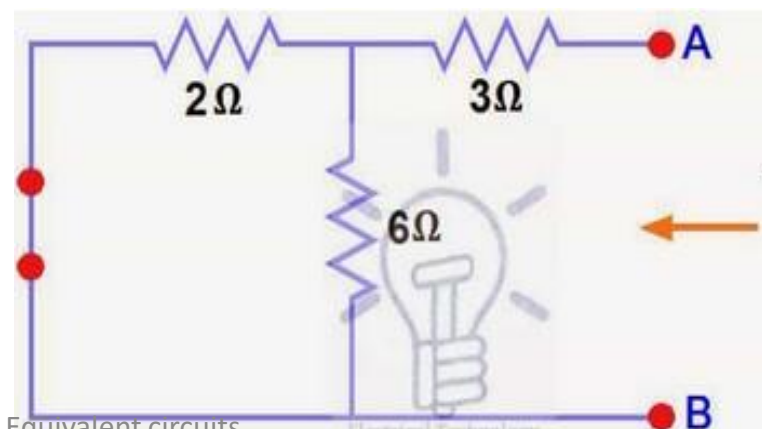
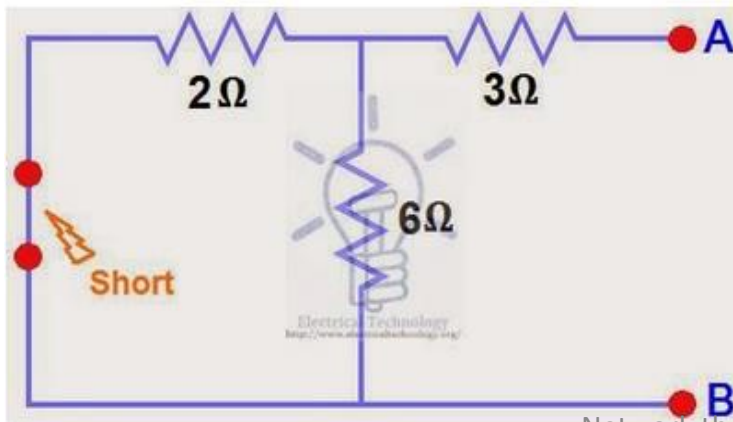
$$I_{SC} = I_N = 3A \times [(6\Omega \div (3\Omega + 6\Omega))] = 2A.$$

$$I_{SC} = I_N = 2A.$$

Cont.



- STEP 2.** Open Current Sources, Short Voltage Sources and Open Load Resistor and Calculate /measure the Open Circuit Resistance. This is the **Norton Resistance (R_N)**.



Cont.

- We can see that **3Ω resistor is in series** with a **parallel combination** of **6Ω resistor and 2Ω resistor**. i.e.

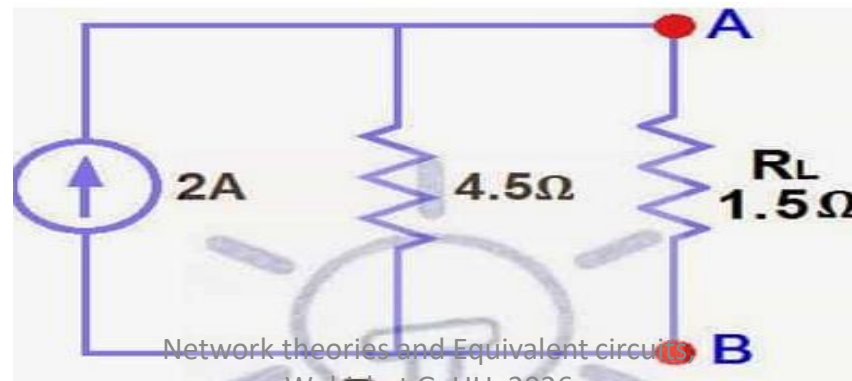
$$3\Omega + (6\Omega \parallel 2\Omega) \dots (\parallel = \text{in parallel with})$$

$$R_N = 3\Omega + [(6\Omega \times 2\Omega) \div (6\Omega + 2\Omega)]$$

$$R_N = 3\Omega + 1.5\Omega$$

$$R_N = 4.5\Omega$$

- STEP 3.** Connect the **R_N in Parallel with I_N** and reconnect the load resistor. This is Norton Equivalent circuit with.



Cont.

$$I_L = I_N \times [R_N \div (R_N + R_L)]$$
$$= 2A \times (4.5\Omega \div 4.5\Omega + 1.5\Omega) \rightarrow = 1.5A$$

$$I_L = 1.5A$$

Load Voltage across Load Resistor...

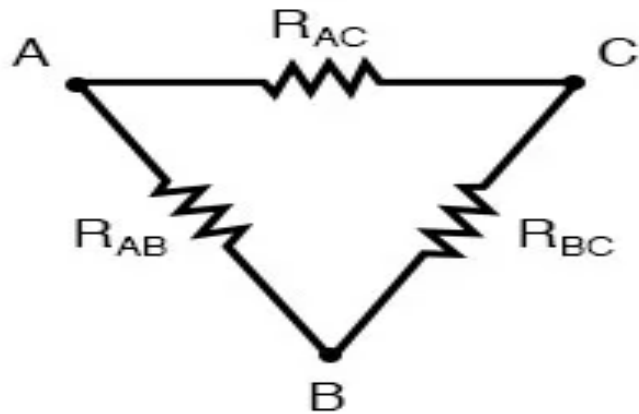
$$V_L = I_L \times R_L$$
$$V_L = 1.5A \times 1.5\Omega$$
$$V_L = 2.25V$$

Δ -Y and Y- Δ Conversions

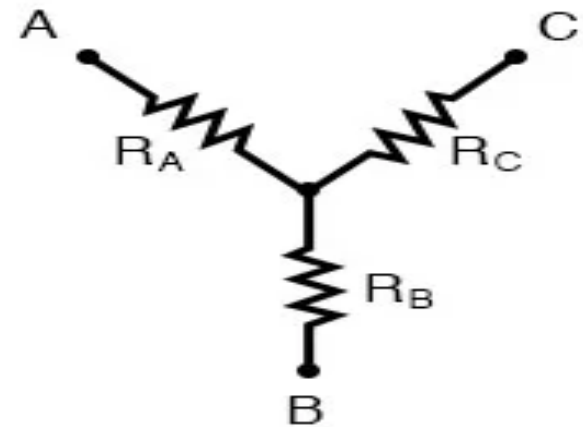
- What if resistors are neither in series nor in parallel?. **Delta**-**Wye**.
- Many circuits can be simplified using three-terminal equivalent networks. These are of the type (Y) wye or tee (T) and the delta (Δ) or pi (Π) networks.

Cont.

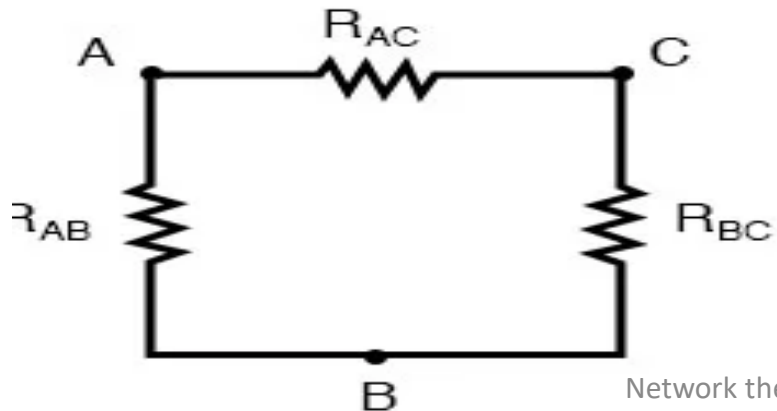
Delta (Δ) network



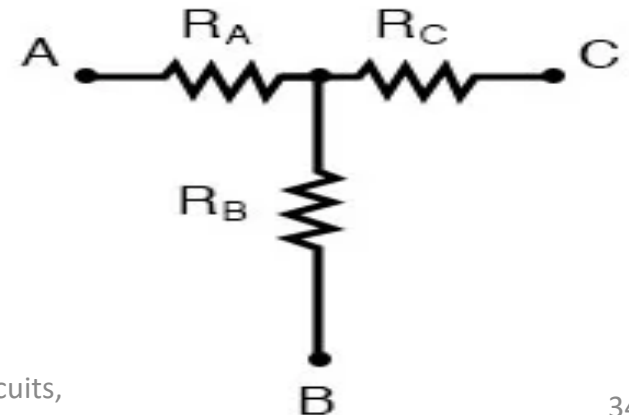
Wye (Y) network



Pi (π) network



Tee (T) network



Cont.

- There are several equations used to convert one network to the other:

To convert a Delta (Δ) to a Wye (Y)

$$R_A = \frac{R_{AB}R_{AC}}{R_{AB} + R_{AC} + R_{BC}}$$

$$R_B = \frac{R_{AB}R_{BC}}{R_{AB} + R_{AC} + R_{BC}}$$

$$R_C = \frac{R_{AC}R_{BC}}{R_{AB} + R_{AC} + R_{BC}}$$

To convert a Wye (Y) to a Delta (Δ)

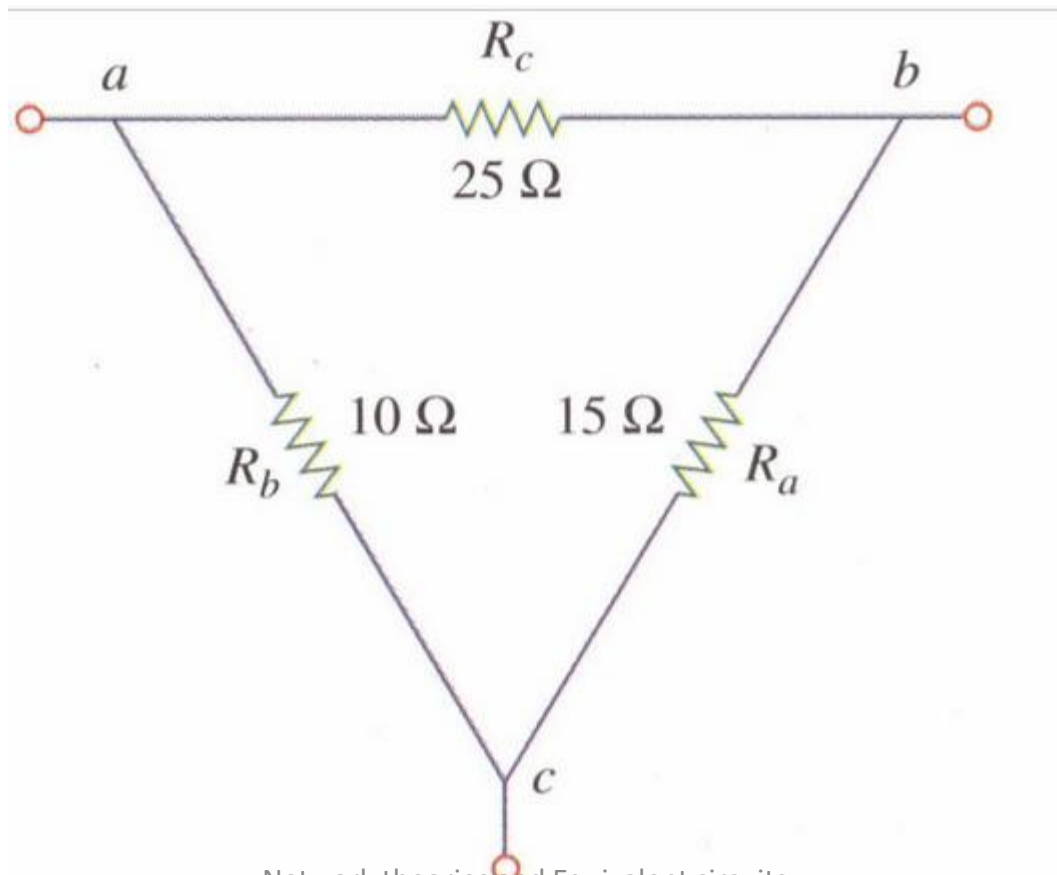
$$R_{AB} = \frac{R_A R_B + R_A R_C + R_B R_C}{R_C}$$

$$R_{BC} = \frac{R_A R_B + R_A R_C + R_B R_C}{R_A}$$

$$R_{AC} = \frac{R_A R_B + R_A R_C + R_B R_C}{R_B}$$

Cont.

- **Example 1:-** Transform the delta network to its equivalent wye network.

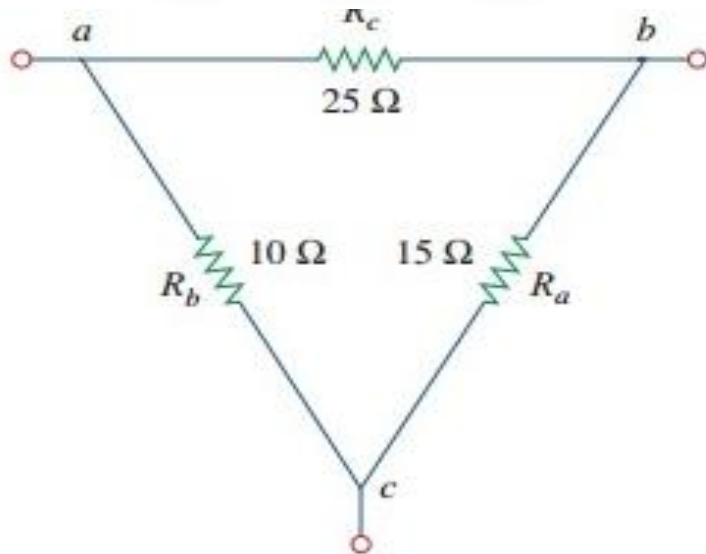


Cont.

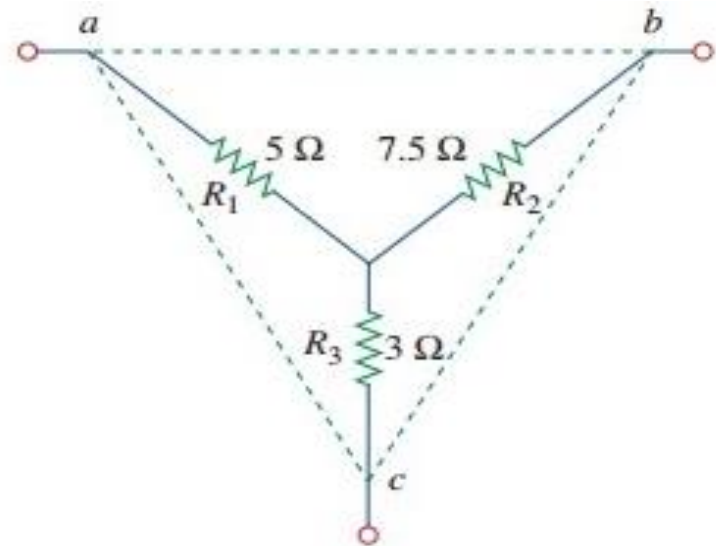
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{15 + 10 + 25} = \frac{250}{50} = 5 \Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{25 \times 15}{50} = 7.5 \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{50} = 3 \Omega$$



(a)



(b)